



Name: Kakumanu Rajesh Kumar

Role: Senior Full Stack Developer (Node.js/NestJS/AWS)

Total Experience: 8 years

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Working At: TAO Digital Solutions

SUMMARY

Senior Software Engineer with 8+ years of experience specializing in high-availability backends (**Node.js/NestJS, Python/FastAPI**) and **AWS cloud architectures**. Expert in AI-Native Development, combining a deep foundation in **Algorithms, Data Structures, and Deep Learning (RNNs)** with next-gen engineering toolkits (**LangChain, LangGraph, Claude Code, GitHub Copilot**) to drive a 2x increase in engineering velocity. Driven by a hands-on, "learning-by-doing" methodology, consistently mastering emerging paradigms like the **Model Context Protocol (MCP)** through rapid prototyping. Proven track record of architecting secure, event-driven systems across healthcare, e-commerce, and automotive sectors. Adept at modernizing legacy infrastructures, leading cross-functional teams, and delivering scalable code within high-velocity Agile environments.

TECHNICAL PROFICIENCIES

- **AI & Agentic Engineering:** Model Context Protocol (MCP), GitHub Copilot (Agent Mode), Claude Code, LangChain, LangGraph, Agentic Workflows, AI-Augmented Refactoring, Context and Prompt Engineering for SDLC.
- **Backend & Frameworks:** Node.js, NestJS, Express.js, Python (FastAPI, Flask, Django), GraphQL, Microservices Architecture, Event-Driven Design.
- **Mathematical & Deep Learning Stack:** NumPy (Vectorized Computations), Scikit-Learn (Preprocessing & Evaluation), Matplotlib & Seaborn (Data Visualization), PyTorch / TensorFlow (RNN, LSTM, GRU Implementation), Pandas (Data Manipulation).
- **Core AI/ML Algorithms:** Recurrent Neural Networks (RNNs), Backpropagation Through Time (BPTT), Gradient Descent variants (Adam, RMSprop), Feature Scaling, Supervised Learning Foundations (Regression, Classification).
- **Core Data Structures & Complexity:** Tensors (N-Dimensional Arrays), Computation Graphs, Multidimensional Matrices, Hash Tables/Dictionaries, Directed Acyclic Graphs (DAGs for workflow execution), Big O Time/Space Complexity.
- **Automated Testing & QA:** Playwright MCP (AI-Driven Testing), Jest, Mocha, PyTest, Self-healing Locators, Automated Unit Test Generation.

- **Frontend & UI:** Next.js, React.js, TypeScript, Figma-to-Code Automation (MCP), Tailwind CSS, Responsive Design, State Management (Redux/Context API).
 - **Cloud (AWS):** Lambda (Serverless), Cognito, SNS, SES, Secrets Manager, S3, EC2, API Gateway, CodePipeline (CI/CD).
 - **Databases & Caching:** PostgreSQL, MySQL, MongoDB, DynamoDB, SAP HANA, Redis (Distributed Caching).
 - **APIs & Security:** RESTful APIs, Swagger/OpenAPI, OAuth 2.0, Role-Based Access Control (RBAC).
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PROFESSIONAL EXPERIENCE

TAO Digital Solutions | Senior Software Engineer | Oct 2023 – Present

AI Research & Core Engineering (Internal R&D) | Apr 2026 – Present

- **Multi-Agent Runtimes:** Architecting stateful, multi-agent execution frameworks using **LangGraph** and **LangChain** to prototype autonomous task-routing, dynamic context-stitching, and self-correcting agent loops.

Project: Neuron Performance Suite (NPS) | Client: CDK Global Stack: *Next.js, NestJS, TypeScript, PostgreSQL, AWS, GitHub Copilot (Agent Mode), MCP*

- **Agentic Development & MCP Integration:** Pioneered the use of **Model Context Protocol (MCP)** to bridge AI agents with internal development tools. Implemented **Figma MCP servers** to automate the conversion of high-fidelity designs into production-ready **Next.js** components, reducing frontend scaffolding time by **50%**.
- **AI-Driven Quality Assurance:** Leveraged **Playwright MCP** to transform testing into an autonomous process; utilized AI agents to reason through live DOM states and accessibility trees, creating self-healing end-to-end tests that improved test stability and coverage by **35%**.
- **Agentic Workflow Orchestration:** Deployed **GitHub Copilot (Agent Mode)** for multi-step refactoring of legacy modules and autonomous generation of complex **Jest** test suites, accelerating feature velocity by **25%**.
- **Architected Microservices:** Engineered a modular NestJS architecture using AI-augmented prototyping to document internal service-to-service communication protocols 3x faster than traditional methods.
- **Automated Communication Engine:** Integrated **AWS SNS/SES** to build a self-healing notification system, eliminating 100% of manual distribution intervention through agentic error-monitoring scripts.

Project: CareExpand (EMR Platform) Stack: *NestJS, TypeScript, MySQL, Sequelize ORM, Microservices*

- **EMR Modernization:** Directed the migration of legacy clinical systems to a NestJS microservices framework, reducing technical debt and increasing system uptime.
- **Data Integrity:** Architected complex relational schemas in MySQL for patient records, ensuring strict adherence to healthcare data standards.

Project: *twiist™ AID System* | **Client:** *Sequel Stack: NestJS, Sequelize, AWS Lambda, SNS, SES, Cognito*

- **Healthcare Integration:** Developed an orchestration portal linking **Tidepool, Salesforce, and Mulesoft**, managing critical data for insulin delivery devices.
 - **Compliance:** Built HIPAA-compliant portals for PwDs (Persons with Diabetes), ensuring secure, real-time glycemic data monitoring.
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Mobile Programming Pvt. Ltd. | Software Developer | Sept 2021 – Sept 2023

Project: *Core E-commerce Microservices* | **Client:** *CaratLane Stack: Node.js, Express, GraphQL, Redis, MySQL, AWS Redshift, Elasticsearch*

- **Search Optimization:** Tuned Elasticsearch for a high-traffic jewelry platform, significantly improving product discovery speed and conversion rates.
- **Data Orchestration:** Developed a GraphQL layer to aggregate data from multiple services, reducing frontend over-fetching by 40%.
- **Analytics Pipeline:** Architected data sync pipelines from MySQL to AWS Redshift for real-time business intelligence.

Project: *SAP Value Lifecycle Manager (VLM)* | **Client:** *SAP Stack: Node.js, Express, SAP HANA DB*

- **Enterprise Logic:** Engineered HANA DB stored procedures to automate complex value-realization calculations for global enterprise clients.

Project: *Smart Restaurant System (POS)* *Stack: Node.js, Express, Parse Server, MongoDB*

- **BaaS Implementation:** Leveraged Parse Server to rapidly deploy a secure, real-time POS system for high-volume restaurant environments.
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Consortium of Institutions of Higher Learning | Software Developer | July 2017 – Aug 2021

- Developed web applications using Node.js and JavaScript to streamline institutional workflows.
 - Implemented secure authentication and optimized database queries for student management
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systems.

GoLive Gaming Solutions Pvt. Ltd. | Game Developer | Sep 2016 – Apr 2017

- Developed interactive game logic and backend services using JavaScript and associated frameworks.
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EDUCATION

- **Master of Technology (IT)** | JNTUH, Hyderabad | 2015 – 2017
- **Bachelor of Technology (Civil Engineering)** | St. Mary's Group of Institutions | 2010 – 2014